
PH102: Physics-2

Electrodynamics

Lecture # 10

Electrostatics – Electric fields in
matter – Polarization and bound
charges

08.02.2024

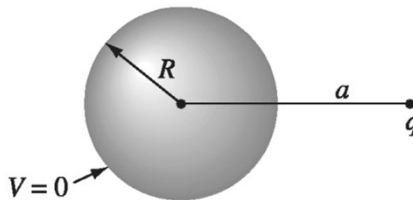
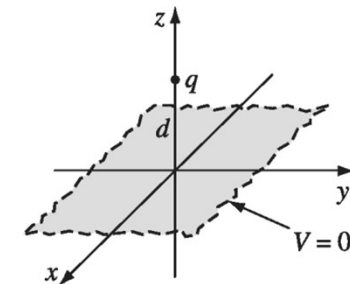
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Previous lecture

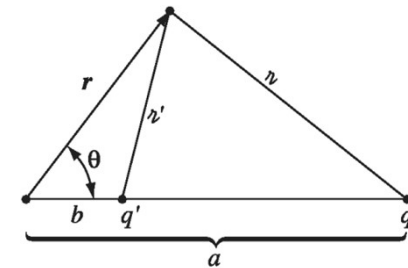
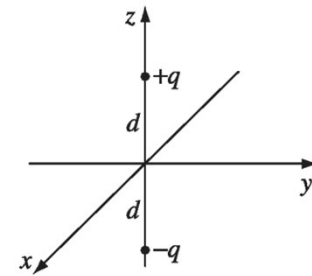
➤ Method of images:

Uniqueness theorem .

Original problem



Equivalent problem



➤ Method of separation of variables: $V(x, y, z) = X(x)Y(y)Z(z)$.

$$V(x, y) = (Ae^{kx} + Be^{-k})(C \sin ky + D \cos ky)$$

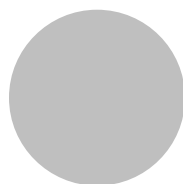
$$C_n = \frac{a}{2} \int_0^a V_o(y) \sin \left(\frac{n' \pi y}{a} \right) dy$$

Dielectrics

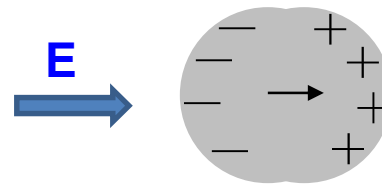
- Most of everyday objects with some approximation belong to one of two large classes: **conductors and insulators (or dielectrics)**.
- **Conductors : Substances contain an “unlimited” supply of charges that are free to move about through the material.**
- **Dielectrics : All charges are attached to specific atoms or molecules. All they can do is move a bit within the atom or molecule.**
- **In dielectrics, microscopic displacements are not as dramatics as the wholesale rearrangement of charge carriers in conductor, but their cumulative effects account for the characteristic behavior of dielectric materials.**
- There are actually two principal mechanisms by which electric fields can distort the charge distribution of a dielectric atom or molecule: **stretching and rotating.**

Induced dipoles

What happens to a neutral atom when it is placed in an electric field E ?



Neutral atom: Positive and negative charge centers coincide in the absence of an external electric field.



Positive and negative charge centers do not coincide in the presence of an external electric field E .

- **Positive charge center** (nucleus) moves along the direction of E .
- **Negative charge center** (electron cloud) moves in the direction opposite to E .
- The two opposing forces - E pulling the electrons and nucleus apart, their mutual attraction drawing them back together - reach a balance, and **the atom is polarized**.

The atom or molecule acquires a tiny dipole moment p , along the direction of E and proportional to it.

$$p = \alpha E.$$

H	He	Li	Be	C	Ne	Na	Ar	K	Cs
0.667	0.205	24.3	5.60	1.67	0.396	24.1	1.64	43.4	59.4

Experimentally determined atomic polarizabilities ($\frac{\alpha}{4\pi\epsilon_0}$) in the unit of 10^{-30} m^3 .

(α is the atomic polarizability; ease with which an atom can be polarized)

Polarizability of molecules

For CO₂ molecule, α is different in different directions.

When the field is at some angle to the axis of the molecule, we write

$$\mathbf{p} = \alpha_{\perp} \mathbf{E}_{\perp} + \alpha_{\parallel} \mathbf{E}_{\parallel}$$

Thus, the induced dipole moment may not even be in the same direction as $\mathbf{E}$.

For a completely asymmetrical molecule,

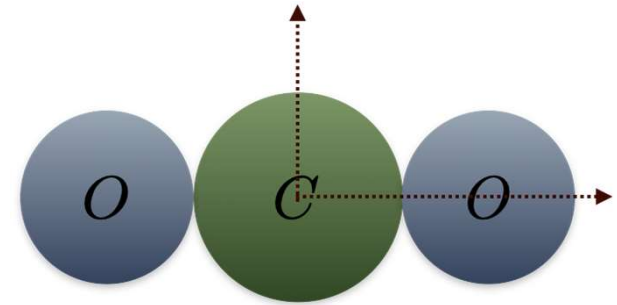
$$p_x = \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z$$

$$p_y = \alpha_{yx}E_x + \alpha_{yy}E_y + \alpha_{yz}E_z$$

$$p_z = \alpha_{zx}E_x + \alpha_{zy}E_y + \alpha_{zz}E_z$$

The values of α_{ij} can always be chosen along the "principal" axes such that all the off-diagonal terms (α_{xy} , α_{zx} , etc.) vanish, leaving just three nonzero polarizabilities: α_{xx} , α_{yy} , and α_{zz} .

$$\alpha = 2 \times 10^{-40} \text{ C}^2 \cdot \text{m/N}$$

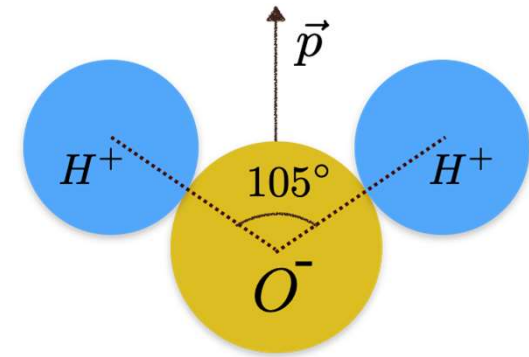


$$\alpha = 4.5 \times 10^{-40} \text{ C}^2 \cdot \text{m/N}$$

The set of nine constants α_{ij} constitute the **polarizability tensor** for the molecule.

Alignment of polar molecules

- The neutral atom had no dipole moment to start with, $\vec{p}$ was induced by the applied field $\vec{E}$.
- Some molecules have *built-in, permanent dipole moment* (polar molecules).



In the water molecule the electrons tend to cluster around the oxygen atom.

The molecule is bent at 105° and so leaves a negative charge at the vertex and a net positive charge at the opposite end.

$p = 6.1 \times 10^{-30}$ Cm for water molecule is unusually large and accounts for its effectiveness as a solvent.

What happens when polar molecules are placed in an electric field?

Torque on a permanent dipole in uniform field

$\mathbf{F}_+ = q\mathbf{E}$ cancel $\mathbf{F}_- = -q\mathbf{E}$ but there is a net torque,

$$\begin{aligned}\mathbf{N} &= (\mathbf{r}_+ \times \mathbf{F}_+) + (\mathbf{r}_- \times \mathbf{F}_-) \\ &= [(\mathbf{d}/2) \times (q\mathbf{E})] + [(-\mathbf{d}/2) \times (-q\mathbf{E})] \\ &= q\mathbf{d} \times \mathbf{E}.\end{aligned}$$

A dipole $\mathbf{p} = q\mathbf{d}$ in the uniform field experiences a torque

$$\mathbf{N} = \mathbf{p} \times \mathbf{E}$$

$\mathbf{N}$ is in such a direction as to line $\mathbf{p}$ up parallel to $\mathbf{E}$.

What if $\mathbf{E}$ is non uniform?

There will be a net force on the dipole, in addition to the torque,

$$\mathbf{F} = \mathbf{F}_+ + \mathbf{F}_- = q(\mathbf{E}_+ - \mathbf{E}_-) = q(\Delta\mathbf{E})$$

For a very short dipole, small change in E_x is

$$\Delta E_x = \frac{\partial E_x}{\partial x} \Delta x + \frac{\partial E_x}{\partial y} \Delta y + \frac{\partial E_x}{\partial z} \Delta z = (\mathbf{d} \cdot \nabla) E_x$$

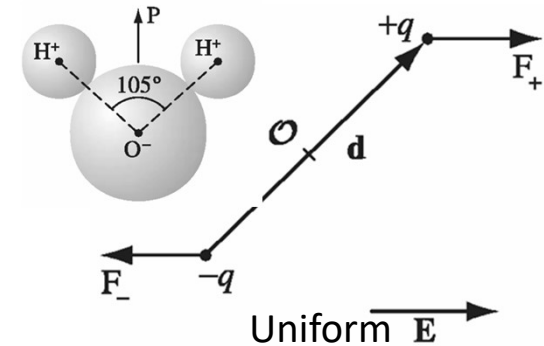
In general,

$$\Delta\mathbf{E} = (\mathbf{d} \cdot \nabla)\mathbf{E}$$

Therefore,

$$\mathbf{F} = q\Delta\mathbf{E} = (\mathbf{p} \cdot \nabla)\mathbf{E}$$

Torque about an arbitrary point is, $\mathbf{N} = (\mathbf{p} \times \mathbf{E}) + (\mathbf{r} \times \mathbf{F})$



Polarization

What happens to a piece of dielectric material when it is placed in an electric field?

Neutral atoms (or non-polar molecules) : Induce tiny dipole moment, pointing in the same direction as the field (**stretching**).

Polar molecules: Experience a torque, tending to line it up along the field direction (**rotating**).

Results : A lot of little dipoles point along the direction of the field - the material becomes **polarized**.

A convenient measure of this of this effect is

$P \equiv$ dipole moment per unit volume,

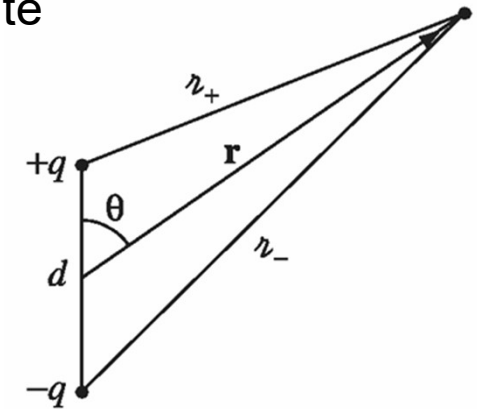
which is called the **polarization**.

Potential at points far from the dipole

A (physical) **electric dipole** consists of two equal and opposite charges ($\pm q$) separated by a distance d .

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_+} - \frac{q}{r_-} \right)$$

Also, $r_{\pm}^2 = r^2 + (d/2)^2 \mp rd \cos \theta = r^2 \left(1 \mp \frac{d}{r} \cos \theta + \frac{d^2}{4r^2} \right)$.



In the regime $r \gg d$, the third term is negligible, and with binomial expansion,

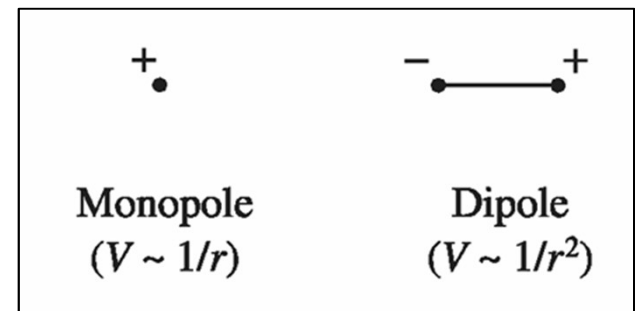
$$\frac{1}{r_{\pm}} \cong \frac{1}{r} \left(1 \mp \frac{d}{r} \cos \theta \right)^{-1/2} \cong \frac{1}{r} \left(1 \pm \frac{d}{2r} \cos \theta \right).$$

or, $\frac{1}{r_+} - \frac{1}{r_-} \cong \frac{d}{r^2} \cos \theta$,

$$V(\mathbf{r}) \cong \frac{1}{4\pi\epsilon_0} \frac{qd \cos \theta}{r^2}$$

Therefore,

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$$



The electric field of the dipole

Electric potential due to the dipole is

$$V_{\text{dip}}(r, \theta) = \frac{\hat{\mathbf{r}} \cdot \mathbf{p}}{4\pi\epsilon_0 r^2} = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}.$$

The negative gradient of V gives the electric field,

$$E_r = -\frac{\partial V}{\partial r} = \frac{2p \cos \theta}{4\pi\epsilon_0 r^3},$$

$$E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{p \sin \theta}{4\pi\epsilon_0 r^3},$$

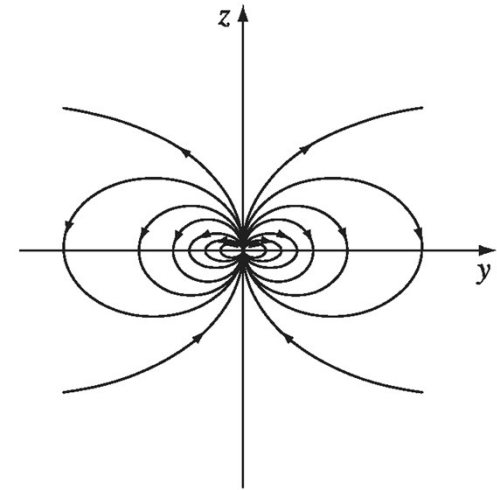
$$E_\phi = -\frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} = 0.$$

Therefore,

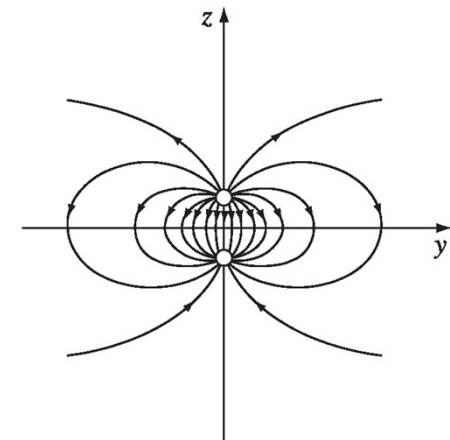
$$\mathbf{E}_{\text{dip}}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}})$$

In the coordinate-free form,

$$\mathbf{E}_{\text{dip}}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$$



Perfect dipole



Physical dipole

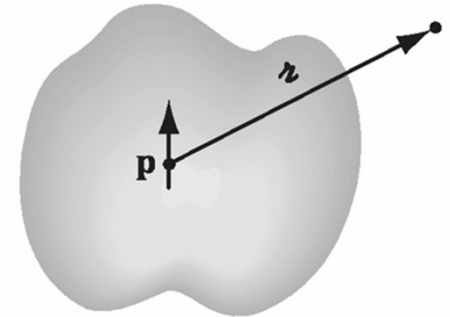
The field of a polarized object

Consider a piece of polarized material with polarization $\mathbf{P}$.

We know the field of an individual dipole, so chop the material up into infinitesimal dipoles and integrate to get the total.

For a single dipole $\mathbf{p}$, $V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}$

Since $\mathbf{p} = \mathbf{P}d\tau'$, the total potential is



$$\begin{aligned}
 V(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \int_V \frac{\mathbf{P}(\mathbf{r}') \cdot \hat{\mathbf{r}}}{r^2} d\tau' = \frac{1}{4\pi\epsilon_0} \int_V \mathbf{P} \cdot \nabla' \left(\frac{1}{r} \right) d\tau'. & \left(\nabla' \left(\frac{1}{r} \right) = \frac{\hat{\mathbf{r}}}{r^2} \right) \\
 &= \frac{1}{4\pi\epsilon_0} \left[\int_V \nabla' \cdot \left(\frac{\mathbf{P}}{r} \right) d\tau' - \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau' \right] & \boxed{\nabla \cdot (f\mathbf{A}) = f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f)} \\
 &= \frac{1}{4\pi\epsilon_0} \oint_S \frac{1}{r} \mathbf{P} \cdot d\mathbf{a}' - \frac{1}{4\pi\epsilon_0} \int_V \frac{1}{r} (\nabla' \cdot \mathbf{P}) d\tau'. & \text{(invoking the divergence theorem)}
 \end{aligned}$$

The first term looks like the potential of a surface charge, $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$
 ($\hat{\mathbf{n}}$ is the normal unit vector)

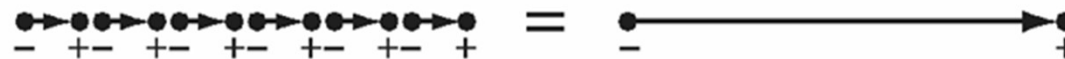
The second term looks like the potential of a volume charge, $\rho_b = -\nabla \cdot \mathbf{P}$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \oint_S \frac{\sigma_b}{r} da' + \frac{1}{4\pi\epsilon_0} \int_v \frac{\rho_b}{r} d\tau'$$

What this means is that the potential (and hence also the field) of a polarized object is the same as that **produced by a volume charge density plus a surface charge density**. **These volume and surface charges are referred to as bound charges.**

Physical interpretation of bound charges

In a long string of dipoles, as shown along the line, the head of one effectively cancels the tail of its neighbor, but at the ends there are two charges left over: plus at the right end and minus at the left.



The net charge at the ends is a bound charge that cannot be removed.

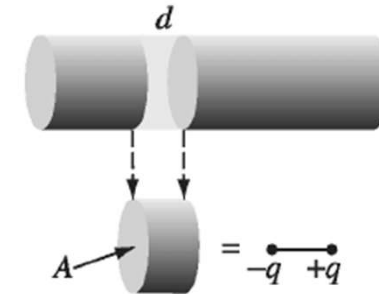
What is the actual **amount of bound charge** resulting from a given polarization?

Consider a "tube" of dielectric parallel to $\mathbf{P}$.

For the tiny chunk, $p = P(Ad) = (PA)d = qd$.

Therefore, $\sigma_b = q/A = P$.

For an oblique cut, $A = A_{\text{end}} \cos \theta$,

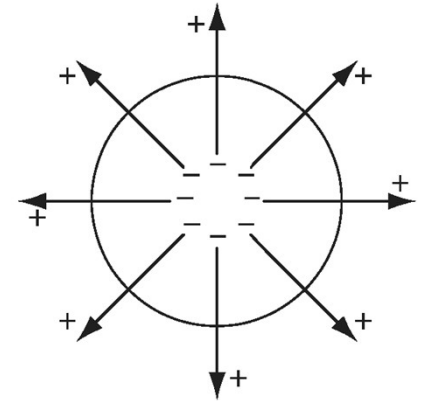


$$\sigma_b = \frac{q}{A_{\text{end}}} = P \cos \theta = \mathbf{P} \cdot \hat{\mathbf{n}}.$$

(Contd...)

If the polarization is non-uniform, we get accumulations of bound charge *within* the material, as well as on the surface.

A diverging $\mathbf{P}$ results in a pileup of negative charge. The net bound charge in a given volume is equal and opposite to the amount that has been pushed out through the surface.



$$\int_V \rho_b d\tau = - \oint_S \mathbf{P} \cdot d\mathbf{a} = - \int_V (\nabla \cdot \mathbf{P}) d\tau. \Rightarrow \rho_b = -\nabla \cdot \mathbf{P}$$

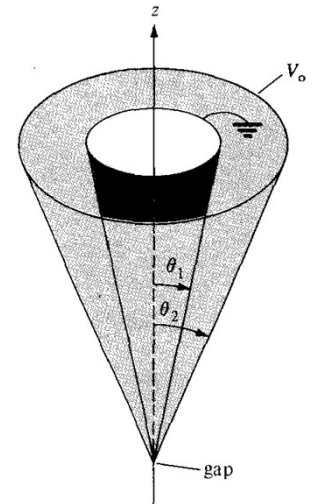
Solved example 1

Two conducting cones ($\theta = \pi/10$ and $\theta = \pi/6$) of infinite extent are separated by an infinitesimal gap at $r = 0$. If $V(\theta = \pi/10) = 0$ and $V(\theta = \pi/6) = 50$ V, find V between the cones.

$$\nabla^2 V = \frac{1}{r^2 \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{dV}{d\theta} \right] = 0$$

$$\frac{d}{d\theta} \left[\sin \theta \frac{dV}{d\theta} \right] = 0 \Rightarrow \sin \theta \frac{dV}{d\theta} = A$$

$$\begin{aligned} V &= A \int \frac{d\theta}{\sin \theta} = A \int \frac{d\theta}{2 \cos \theta/2 \sin \theta/2} = A \int \frac{1/2 \sec^2 \theta/2 d\theta}{\tan \theta/2} = A \int \frac{d(\tan \theta/2)}{\tan \theta/2} \\ &= A \ln (\tan \theta/2) + B \end{aligned}$$



Applying boundary conditions,

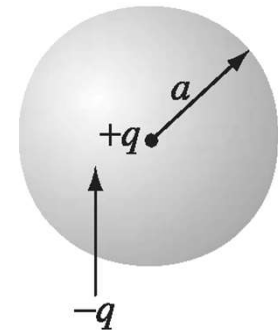
$$V(\theta = \theta_1) = 0 \rightarrow 0 = A \ln (\tan \theta_1/2) + B \Rightarrow B = -A \ln (\tan \theta_1/2) \Rightarrow V = A \ln \left[\frac{\tan \theta/2}{\tan \theta_1/2} \right]$$

$$V(\theta = \theta_2) = V_0 \rightarrow V_0 = A \ln \left[\frac{\tan \theta_2/2}{\tan \theta_1/2} \right] \Rightarrow A = \frac{V_0}{\ln \left[\frac{\tan \theta_2/2}{\tan \theta_1/2} \right]} \Rightarrow V = \frac{V_0 \ln \left[\frac{\tan \theta/2}{\tan \theta_1/2} \right]}{\ln \left[\frac{\tan \theta_2/2}{\tan \theta_1/2} \right]} = \frac{50 \ln \left[\frac{\tan \theta/2}{\tan \pi/20} \right]}{\ln \left[\frac{\tan \pi/12}{\tan \pi/20} \right]}$$

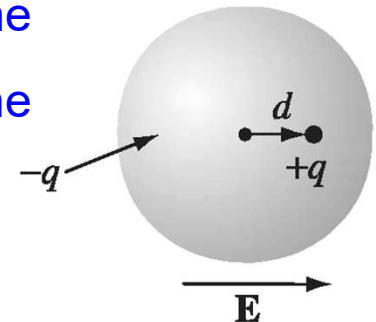
Solved example 2

A primitive model for an atom consists of a point nucleus (+ q) surrounded by a uniformly charged spherical cloud (- q) of radius a . Calculate the atomic polarizability of such an atom.

- The actual displacements involved are extremely small.
- Assume that the electron cloud retains its spherical shape.
- Equilibrium occurs when the nucleus is displaced a distance d from the center of the sphere.



External field pushing the nucleus to the right exactly balances the internal field pulling it to the left: $\mathbf{E} = \mathbf{E}_e$ (field produced by the electron cloud).



$$E_e = \frac{\rho d}{3\epsilon_0} = \frac{q}{\frac{4}{3}\pi a^3} \frac{d}{3\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} \quad \text{and at equilibrium,}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{qd}{a^3} \quad \text{or} \quad p = qd = (4\pi\epsilon_0 a^3) \mathbf{E} \Rightarrow \alpha = 4\pi\epsilon_0 a^3 = 3\epsilon_0 v$$

(v is the volume of the atom.)

Solved example 3

Find the electric field produced by a uniformly polarized sphere of radius R .

Let us first find out the bound charges,

$$\rho_b = -\nabla \cdot \mathbf{P} = 0 \quad (\mathbf{P} \text{ is uniform.})$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}} = P \cos \theta$$

$$V(r, \theta) = \begin{cases} \frac{P}{3\epsilon_0} r \cos \theta, & \text{for } r \leq R, \\ \frac{P}{3\epsilon_0} \frac{R^3}{r^2} \cos \theta, & \text{for } r \geq R. \end{cases}$$

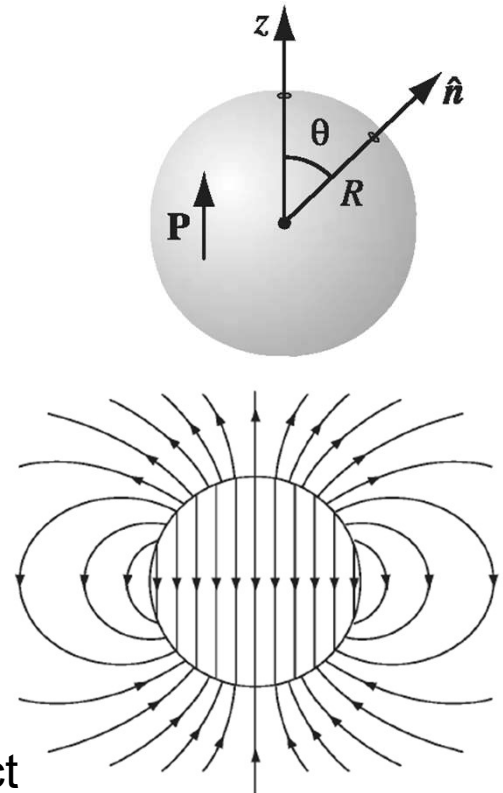
Since $r \cos \theta = z$, the field inside the sphere is uniform,

$$\mathbf{E} = -\nabla V = -\frac{P}{3\epsilon_0} \hat{\mathbf{z}} = -\frac{1}{3\epsilon_0} \mathbf{P}, \quad \text{for } r < R.$$

Outside the sphere the potential is identical to that of a perfect dipole at the origin,

$$V = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}, \quad \text{for } r \geq R, \quad \text{with dipole moment, } \mathbf{p} = \frac{4}{3}\pi R^3 \mathbf{P}.$$

$$\mathbf{E}_{\text{dip}}(r, \theta) = \frac{P}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}).$$



(Contd...)

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \oint \frac{\sigma_b}{r} da = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} d\varphi \int_0^\pi \frac{P \cos \theta R^2 \sin \theta d\theta}{\sqrt{R^2 + z^2 - 2Rz \cos \theta}} = \frac{PR^2}{2\epsilon_0} \int_0^\pi \frac{\cos \theta \sin \theta d\theta}{\sqrt{R^2 + z^2 - 2Rz \cos \theta}}$$

Let $\cos \theta = t$

$$\begin{aligned} V(z) &= \frac{P}{8\epsilon_0} \int_{(R+z)^2}^{(R-z)^2} \frac{(y - R^2 - z^2) dy}{z^2 \sqrt{y}} \\ &= \frac{P}{6z^2 \epsilon_0} [(R^2 + z^2)\{|R + z| - |R - z|\} - Rz\{|R + z| + |R - z|\}] \end{aligned}$$

Case I: $z > R$

$$|R + z| = R + z, \quad |R - z| = z - R$$

$$V(z) = \frac{PR^3}{3\epsilon_0 z^2} \quad \mathbf{E}(z) = \frac{2PR^3}{3\epsilon_0 z^3} \hat{\mathbf{z}}$$

Verify with $V(r, \theta)$

Case II: $z < R$

$$|R + z| = R + z, \quad |R - z| = R - z$$

$$V(z) = \frac{Pz}{3\epsilon_0} \quad \mathbf{E}(z) = -\frac{P}{3\epsilon_0} \hat{\mathbf{z}}$$

Verify with $V(r, \theta)$

Questions?